

CHAPTER SEVEN

General Strength Assessment and Improvement

Endurance athletes can't afford to get stronger at the expense of added muscle mass. They've got to carry that muscle around with them and supply it with fuel and oxygen. The following information is designed with this in mind. You will not find any recipe for gaining muscle mass here.

For the lower body, the goal is first to ensure better single leg, hip, and knee stabilization so that you can apply power more effectively through the major joints to propel yourself farther in each stride. Single-leg stability originates mostly in the hips, specifically in the gluteal muscle group. Without adequate strength in this important muscle group, it is common to see unhealthy knee and ankle alignment. At best, this will have a negative effect on movement economy, power output, and stride length and, at worst, this will set you up for overuse injury. If while standing on one leg, your knee makes a side-to-side movement during either knee flexion or extension, you may have some weakness that warrants remedial work best done with basic strength exercises.

Upper-body strength and its role in performance is often overlooked by runners but especially by skimo racers. The shoulders, arms, back, and chest all contribute directly to stride length in both skiing uphill and in running. If you do not think that the upper body and arm swing play a role in running or skimo, here's a simple experiment: Strap your arms to your sides and then try to run or ski quickly. Get it? Even in running, the arm swing transfers a great deal of energy to the legs.

Having a relaxed shoulder girdle that allows a fluid swinging motion of the arms will impart additional power to your legs and result in added stride length,



An unknown runner, Mount Fuji, Japan. Photo: Martina Valmassoi

whether you're running or skiing. Very high strength levels are not important to runners, but the endurance to maintain that arm swing for hours *is* important. When your back and shoulders become tired, your arm swing will suffer—and so will your stride length.

In the following pages we'll show you some well-tested basic exercises and explain how to combine the ones you need into a workout and then how to progress that workout over the course of weeks and months. Do not feel constrained to do only these exercises. They are by no means all-inclusive. Adding or substituting similar exercises will have similar positive results.

Assessment and Basic Exercise Programs for Legs and Hips

It will be informative to assess your basic strengths and weaknesses by testing yourself using a few simple exercises before launching into a strength-training program. Do this by moving through the exercises listed in the routine that follows this introduction. As you assess your strength, keep in mind that some

athletes may be best served by doing unglamorous remedial strength work just as some need to do the remedial, slow running to correct ADS, while others with good basic strength will be best served by using exercises further along the progression to specific strength.

While there are certainly other strength assessments that can be used, we are going to suggest some simple tests to assess knee stability. Weak hip muscles, specifically the gluteus medius, are the culprit in many running overuse injuries to the knee and lower leg. The aspect to monitor in all these exercises is the ability to keep the knee lined up over the toes when viewed from the front. You can face a mirror or you can simply look down to see if your knee moves side to side during these exercises. If your knee moves inward, toward the centerline of your body when you put weight on it, as your knee either bends or straightens, then you have an alignment issue that is very likely related to a hip stability/strength deficiency.

In most cases this is easily fixed by simply improving the lateral hip muscles' strength by using the exercises described below. However, in some cases the services of a professional strength coach will be required. The simple assessments provided below are not a substitute for professional advice.

Air Squat Test (easy)

This is the simplest assessment. Assume a normal squatting position: feet at shoulder width, toes pointed outward slightly, arms held out in front horizontally at shoulder height. Squat down to bring your butt between your heels as low as you can while keeping the spine in a neutral position (straight, not flexed or hyperextended). Watch your knees as you go down and come up out of the squat.

- Do your knees collapse in at all?
- Can you hold your knees outward so that they remain aligned vertically over your toes?

If you cannot maintain knee alignment over your toes, you can stop the test here and start with the Stage 1 strength program. If you can hold them in alignment with conscious effort, go to the next test.

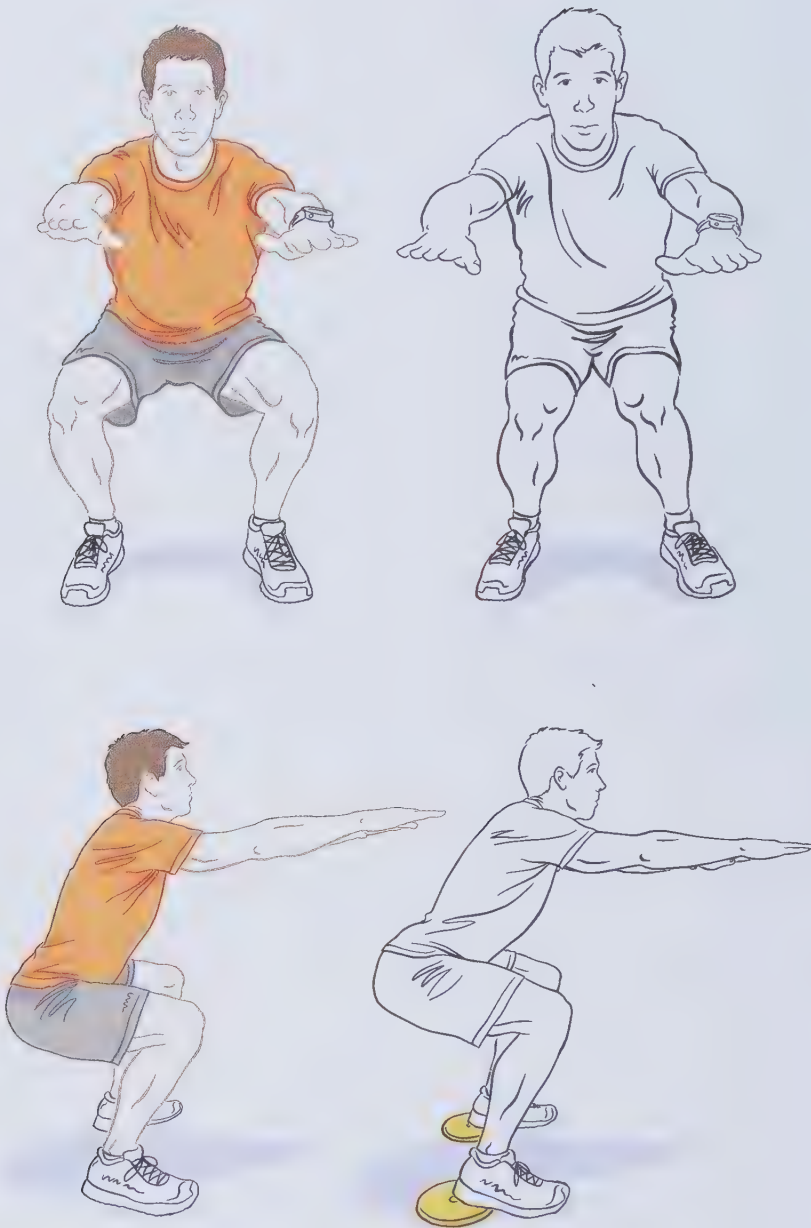


Fig. 7.1 Squat Test

To perform this test stand with feet shoulder width apart and toes angled outward slightly. Squat to where your thighs are parallel to the ground, and stand back up. Do your knees cave in toward your centerline? Or, do they track in line over your toes?

Ryan Sandes looks for more power during a route recon before attempting to break the FKT of the Drakensberg Grand Traverse in Northern Drakensberg, South Africa. Photo: Kelvin Trautman



Box Step Test (medium)

Stand facing a box or step that is just below the height of your knees. Your toes should be no more than twelve inches (thirty centimeters) away from the box. Step up onto the box while watching for any inward collapse of your knee toward the midline of your body. And then step down from the box. Repeat this several times on each leg while observing how your knee tracks.

- How far toward your body's midline does each knee move?
- Do your hips move outward to the side as your knee collapses inward?

If either of your knees collapses to the inside by more than about one inch (2.5 centimeters) and/or your hips shift to the outside by a similar amount while stepping up on the box, you can stop the test here and use the Stage 2 strength program. If you can hold knee alignment, go to the next test.

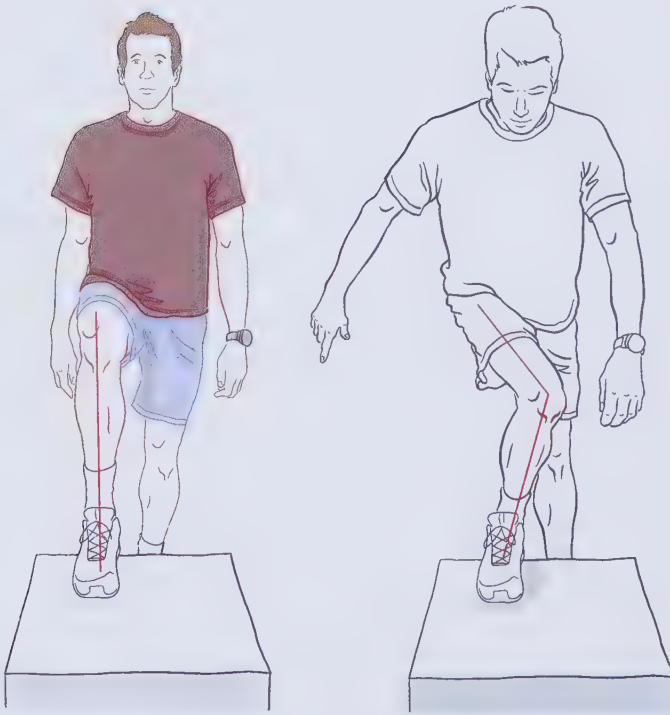


Fig. 7.2 Box Step Test

Stand about 12 inches (30 centimeters) in front of and facing a box or bench that is just below your knee height. Step up and note whether your knee moves inward or if it tracks closely over your toes as you step up.

Sit-to-Stand Test (difficult)

Stand in front of, and facing away from, a hard-surface chair or preferably a bench that allows you to sit on it with your knees bent at ninety degrees and thighs parallel to the ground. Lower yourself to the sitting position on the bench or chair. Do not relax your leg muscles but immediately stand up by using only one leg. Do not hinge your upper body forward at the hips. Try to keep your torso fairly upright. Notice how your knee tracks over the toes of the foot you are standing on as you rise into a standing position. For many people, arising from a seated position using only one leg will be impossible due to strength. In that case, add a thick book or two to the chair or bench so that you can just barely rise on one leg or try the box step test.

- How far toward your body's midline does each knee move?
- Do your hips move outward to the side as the knee collapses inward?

If either of your knees collapses inward by more than one inch (2.5 centimeters), use the Stage 3 strength program. If you can manage this difficult test with good knee tracking, move directly to the specific strength program.

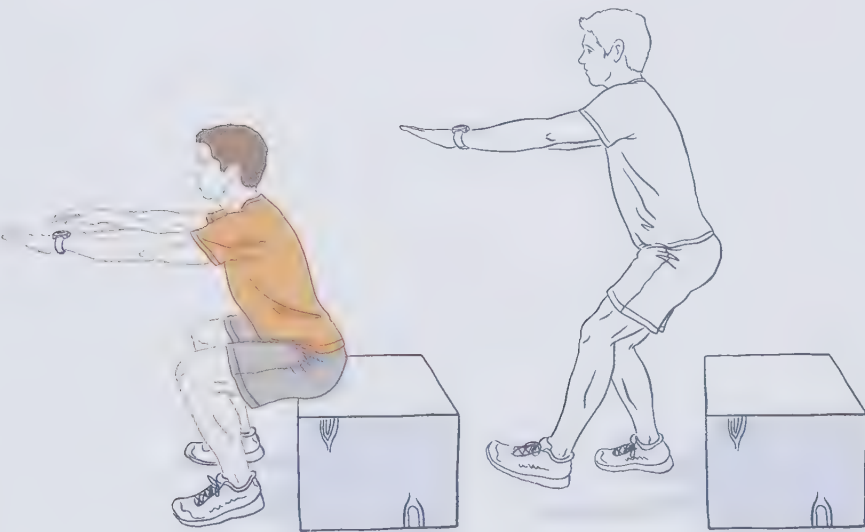


Fig. 7.3 Sit-to-Stand Test

Lower yourself to a sitting position on a bench or chair. Immediately raise yourself back to a standing position using only one leg. This is very challenging to perform at all, let alone with good knee control. The illustrations above show the difference between good form where the knee tracks over the toe and the poor form where the knee caves inward.

If, after three weeks of the following strength programs, you do not see marked improvement in your knee tracking/control, consult with a good physiotherapist to determine if you have some underlying structural issue. When you have achieved the appropriate level of knee control, you can move to the next stage of the strength progression.

Stage 1 Leg and Hip Program

This group of exercises is designed to first teach your brain how to turn on the lateral gluteal muscle groups and then to build some strength in them. Initially use high reps with low resistance; four sets of twenty reps or until you fatigue. Take a one-minute rest between sets. Between sets of these leg exercises, insert a set of upper-body exercises that are described in the next section. Retest for knee alignment after a couple of weeks and move on to Stage 2 when appropriate hip stability is gained.



Fig. 7.4 Squats with Knee Band

A good hip-strengthening exercise can be done by squatting with a rubber band placed just above the knees.

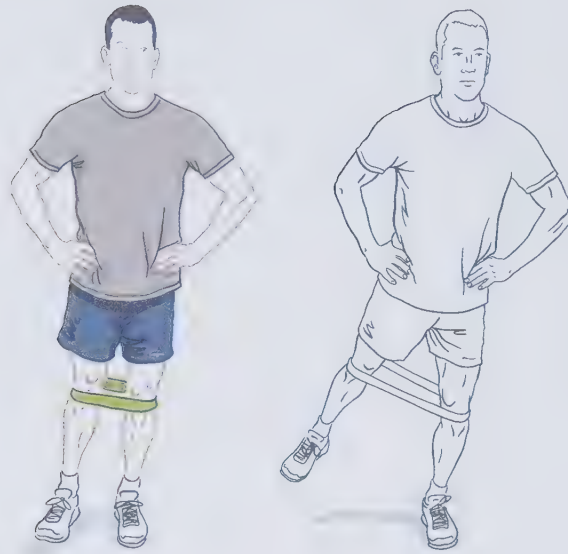


Fig. 7.5 Single Leg Lateral Band

A more advanced hip-strengthening exercise is done by placing a rubber band around your knees (either above or below) and while balanced on one leg, extend the unweighted leg to the side and back at about a 45-degree angle.

Stage 2 Leg and Hip Program

Start these with just your body weight to learn the movements with correct form and control by doing three sets of fifteen reps on each leg with one to two minutes' rest between sets. Progress by increasing reps each session to the point of fatigue. Between sets of these leg exercises, insert a set of upper-body exercises covered in the next section. When you can do three sets of thirty reps with perfect knee control and balance, move to Stage 3.

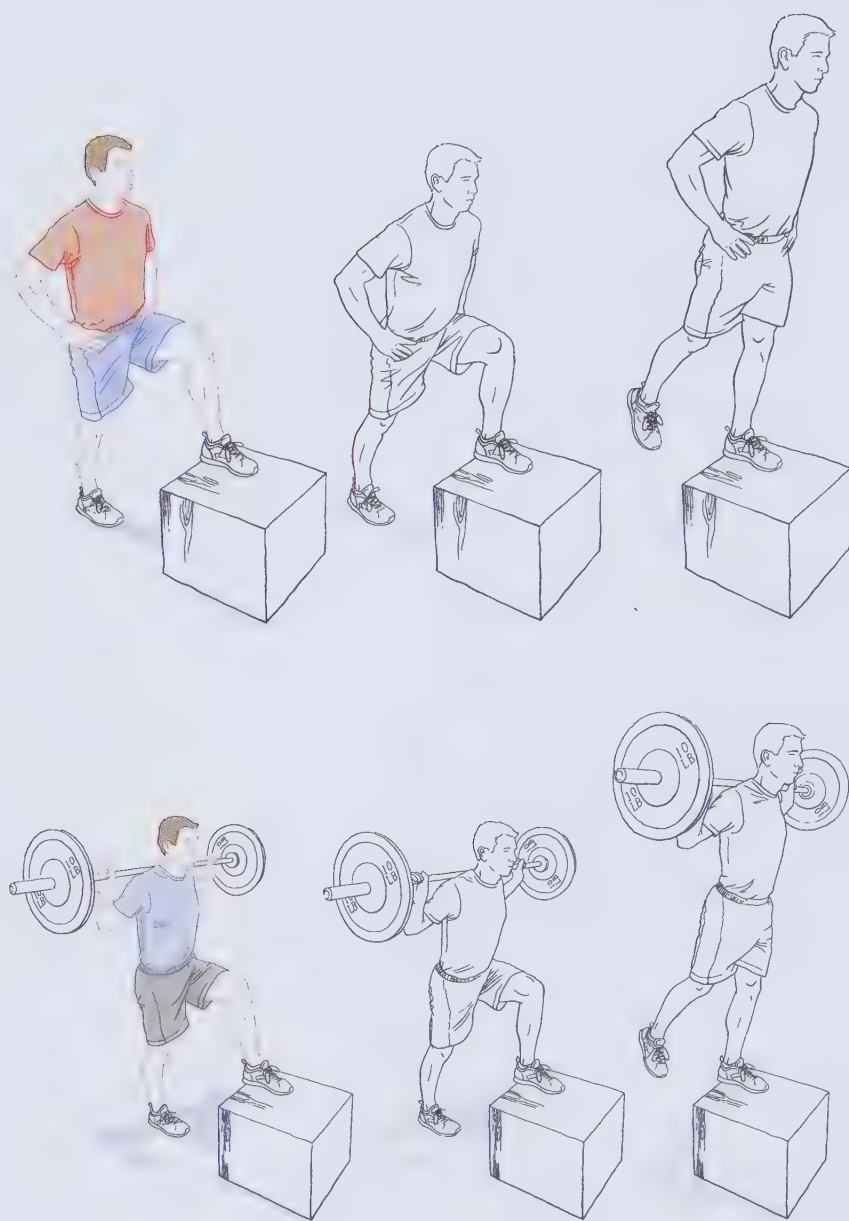


Fig. 8.1 Box Step-Up

Stand about one half of your stride length in front of a box or step that is just below knee height. Step up onto the box mainly using the front leg to pull you up with minimal assist from the rear leg. Maintain proper knee tracking over your toes throughout this exercise.



Fig. 7.7 Box Step-Down

The left figure shows the box step-down exercise with proper knee alignment. The right figure shows the weighted knee collapsing inward due to poor hip strength. If you cannot maintain good knee alignment when doing this exercise, use a lower box to step down from.

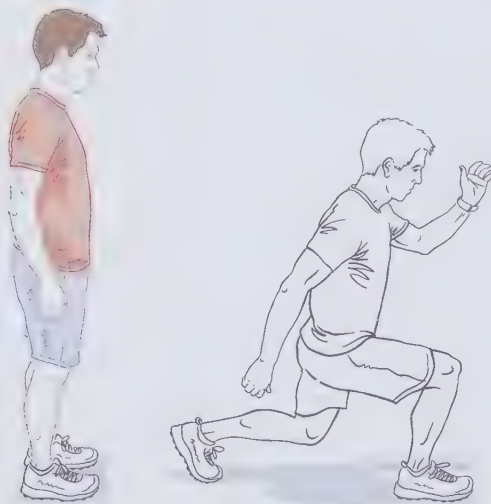


Fig. 7.8 Front Lunge

From a standing position step forward onto a bent front knee. Push back and up to return to the standing position. Alternate legs for the required number of repetitions. As you gain strength you can add resistance by adding a weight vest, holding a barbell to your shoulders, or holding a dumbbell or kettlebell in front of your chest.



Dimitra Teocharis tests her strength at the Marmolada Full Gas race on the Marmolada Glacier in the Val di Fassa, Italy. Photo: Federica Modica

Stage 3 Leg and Hip Program

Once you are proficient with Stage 2 exercises, progress by adding weight to the same exercises as in Stage 2. Select enough resistance in the form of a weight vest, pack, or barbell on your shoulders so that you can just manage eight reps with perfect form. Do four sets of six reps using this weight. Between sets of the leg exercises, insert a set of upper-body exercises covered in the next section. Take a two-minute rest between sets of the same exercise.

Assessment and Basic Exercise Programs for Upper Body

While not particularly obvious for running and skiing, it can still prove very worthwhile for you to evaluate your personal upper-body strengths and weaknesses as well as perhaps to improve them. As with the leg and hip exercises, this list is far from exhaustive and instead represents a bare minimum for those with little to no strength-training background.

Skimo Upper Body

Skimo is a quadrupedal sport: all four limbs are used for locomotion. In skimo's sister sport, cross-country skiing, the arms and upper body play a significant role in propulsion (up to 20 percent in diagonal stride classic technique, which mirrors skimo uphill skinning). Even though the upper body plays a smaller role in skimo due to less ski glide in the climbing sections of the course, there is probably still a 5-10 percent contribution to propulsion from the upper body and arms. Have you noticed how fatigued your arms become during a strenuous climb? This should

be a clue as to their contribution (or lack thereof) to your skinning.

Increasing the strength and endurance of the poling muscles will lead to an improvement in uphill skinning endurance. We will look at the methods a skimo racer can utilize to improve these qualities and show how to incorporate workouts into your training plan a bit later in this section; suffice it to say that if you are not incorporating ski-specific upper-body training into your program, you are not doing everything you could to maximize performance.

Push-Up Assessment

With this single simple test, you can get a good handle on your overall upper-body strength. This is a good exercise for runners because it requires strength in the arms, shoulders, and core. Assume a plank position so that your hands and toes support all your weight as if you were going to do a standard push-up. Keep your spine in a neutral position so that your hips don't sag or rise up into an inverted "V." Place your hands in a comfortable spot below your shoulders.

If you cannot hold this top position for five seconds without sagging or raising your hips, place your hands on a chair or bench top and assume the same push-up position. If you can now manage to hold this top position for five seconds, take a two-minute rest and then count how many push-ups you can do. If you can't hold this top position, then move your hands to a countertop (about one meter high) and try again to complete at least one inclined push-up from your toes. See the illustrations below for clarification. If you can manage to do only inclined push-ups, start with the Stage 1 Upper-Body Program.

If you can hold the normal push-up top position for more than five seconds, rest for two minutes and count the number of push-ups you can do without losing form. If you can do fewer than five proper push-ups, use the Stage 2 Upper-Body Program.

If you can do more than five push-ups in good form, use the Stage 3 Upper-Body Program.

Pull-Up Assessment for Skimo

The simple pull-up (palms facing away) or chin-up (palms facing you) is a good indicator of basic upper-body pulling strength. There are other ways to train ski-specific single-arm poling strength and endurance (to be discussed in the next chapter), but we'll use the pull-up for this general strength assessment.

Using either a chin-up or pull-up grip, hang from a bar and pull yourself up until your chin is just above the bar. Start with slightly bent elbows and with your shoulder blades pulled down (retracted). Don't start in a fully extended hanging position.

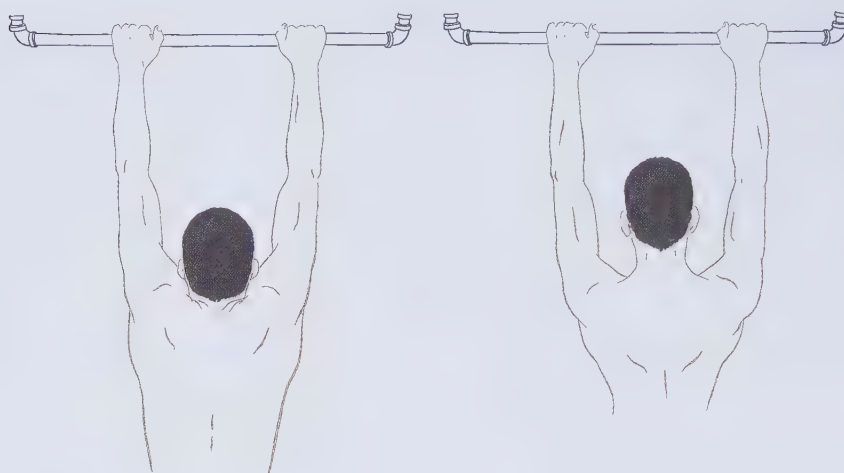


Fig. 7.9 Lower Pull-Up Position

When hanging from a bar at the bottom of the pull-up position try to hang with the shoulder blades retracted as in the right illustration rather than loose and extended as in the left illustration, as this may cause injury to weak shoulders.

If you cannot hang in this bottom position with your shoulders retracted, use the Stage 1 Upper-Body Program.

If you can hang in this lower position, then try hanging with your chin above the bar. Use a stool or have a friend lift you to get to this top position. Hang for as long as you can in this position. If you can hang for two seconds or less, use Stage 1 below. If you can count to three seconds or more, use Stage 2 below.

If you can do between one and four pull-ups, use Stage 3 below.

If you can do five or more pull-ups, use the Specific Strength Training Methods for skimo in chapter eight.



In our view, a well-equipped gym is a simple, clean, space with basic free weights, simple boxes, bars, and rings. Aaron Bollschweiler working on upper-body strength, Momentum Climbing Gym, Sandy, Utah. Photo: Andrew Burr

Stage 1 Upper-Body Program

Runners: Use the appropriate level of incline from the assessment above that allows you to complete three sets of as many inclined push-ups as you can do with two to three minutes' rest between sets, during which you will add a leg exercise. When you can do more than eight push-ups in the third set, progress in this exercise by moving to a lower surface to use for your inclined push-ups and repeat until you can once again do more than eight reps during the third set. Keep decreasing the incline angle while holding good plank form as you get stronger.

Skiers: Add either assisted pull-ups or a lat pull-down machine. Complete three sets of the maximum repetitions you can manage with two to three minutes' rest between sets. When you can do more than eight reps in the third set, increase the weight or reduce the assistance. See the exercise descriptions below.

Inclined Push-Ups

Start with the level of incline that allows you to complete the prescribed set and reps and progress as per the instructions above.

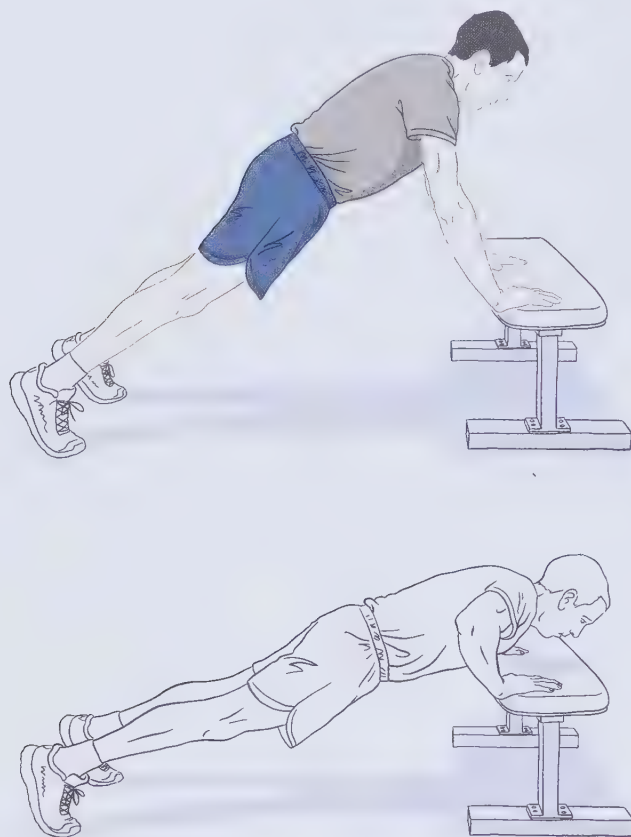


Fig. 7.10 Inclined Push-Up

If you are unable to do a proper push-up on the floor, then elevating your hands on stairs, or even a table, will help you get started in developing important upper-body and core strength.

Assisted Pull-Ups (Skimo)

Use a lat pull-down machine or a pull-up bar and a way for you to remove some body weight like a rubber band or a stool to place a foot on. See illustration for how this exercise is to be performed. Find a level of assistance or the weight that you can barely manage for six repetitions. Use this weight for the prescribed sets and reps. After six workouts like this, try moving to Stage 2.

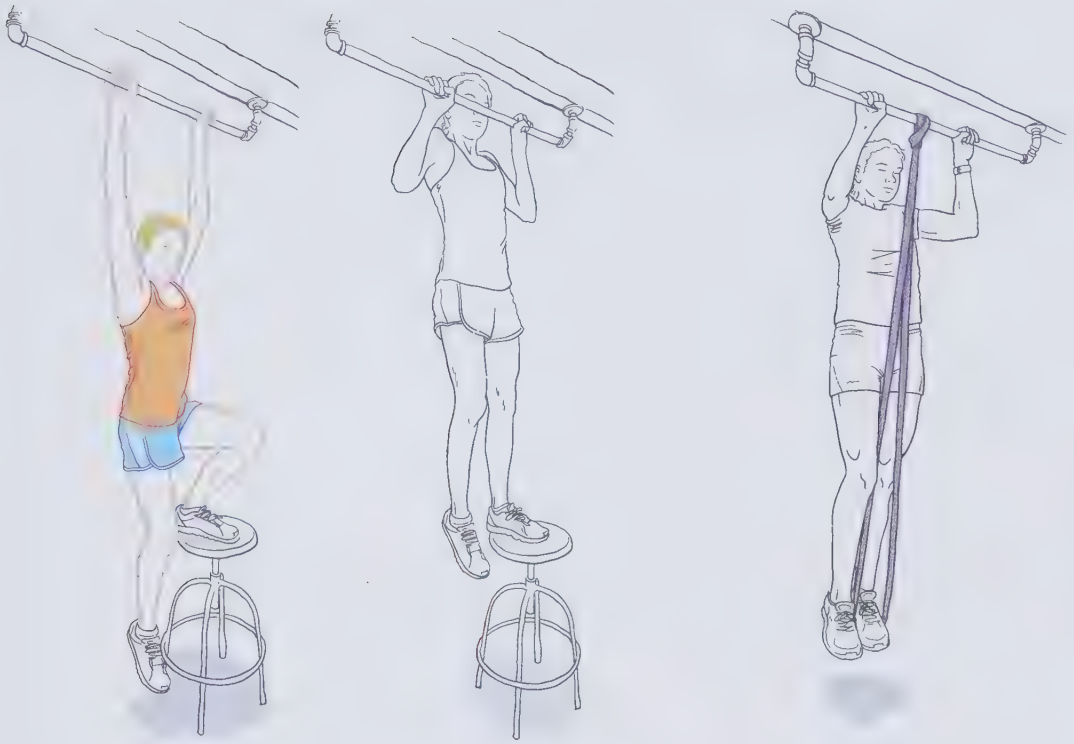


Fig. 7.11 Assisted Pull-Up

If you are unable to perform full body weight pull-ups you can place a foot on a stool, or use a strong rubber band, to give some assistance.

Stage 2 Upper-Body Program

Runners: Once you have mastered the incline push-up by moving to a low incline (such as the first stair tread or a low bench), you will have developed some strength in your arms, shoulders, and core. The next stage will introduce knee push-ups with your weight distributed between your hands, placed on the floor, and your knees. Complete three sets of as many knee push-ups as you can with two minutes' rest between sets, during which you will do a leg exercise. Once you can complete three sets of eight reps of knee push-ups, you should be ready to move on to regular push-ups. Do this by completing as many good-form standard push-ups as you can in each set. In each set, when you can no longer hold good form, switch to knee push-ups, completing the remaining reps up to eight in each set. If you can do more than three sets of eight reps of standard push-ups, move to Stage 3.

Skiers: Using a pull-up bar, get your chin above the bar using assistance. Remove the assistance and hold that top position as long as you can. Drop from the bar and rest for two minutes, during which you will do a leg exercise. Repeat this four times. When you can hold the top position for more than a count of five seconds in the fourth set, use the middle position with elbows bent at ninety degrees and repeat the same sequence.

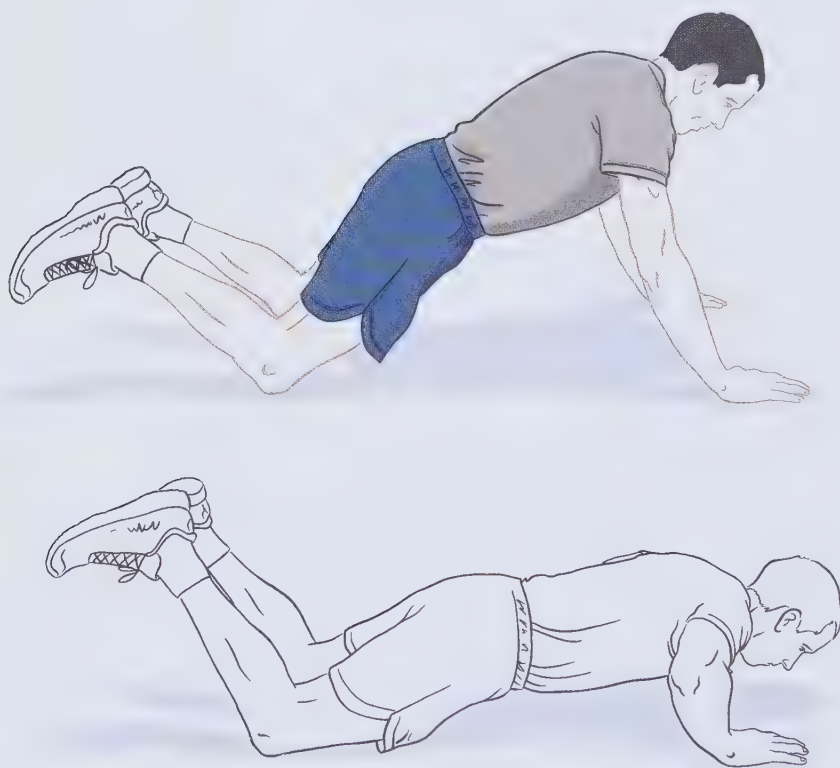


Fig. 7.12 Knee Push-Up

Another good way to work up to full body weight push-ups is to start from your knees. As you progress, you can begin the set on your toes and then drop to your knees when you lose proper form.

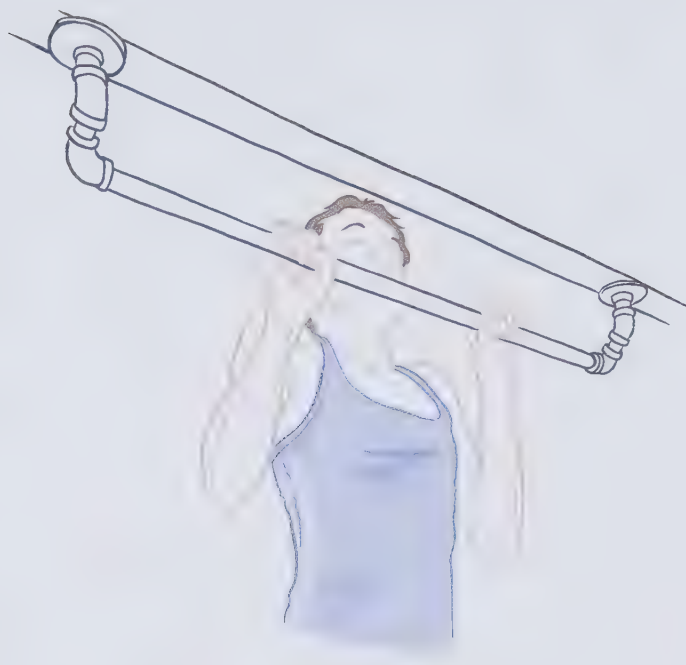


Fig. 7.13 Isometric Bar Hang (Skimo)

Hanging in a locked-off position partway through a pull-up movement is a great exercise for developing strong shoulders. The ability to exert force on your poles when skinning uphill can be a significant benefit for skimo racers.

Stage 3 Upper-Body Program

Runners: Now that you have sufficient strength to do a few good push-ups, develop more strength by making the push-ups more challenging. You'll always start this workout with at least one set of eight regular push-ups followed by three sets adding resistance, limit reps to four per set. Take two to three minutes' rest between sets, during which you will do leg exercises. The easiest way of increasing the resistance is to elevate your feet on a stair, bench, chair, or exercise ball. Wearing a weight vest is another way to add resistance for stronger athletes.

Skiers: By now you are able to do at least one pull-up/chin-up. This routine will build strength quickly by increasing the load and by using the slow-lowering technique. Complete four sets of two to three repetitions with a three-to-five-second slow lower on the last repetition of each set. Add enough extra weight so that you can manage only two to three reps. When you can do five normal pull-ups/chin-ups, then progress to the Specific Strength Training Methods section in chapter eight.

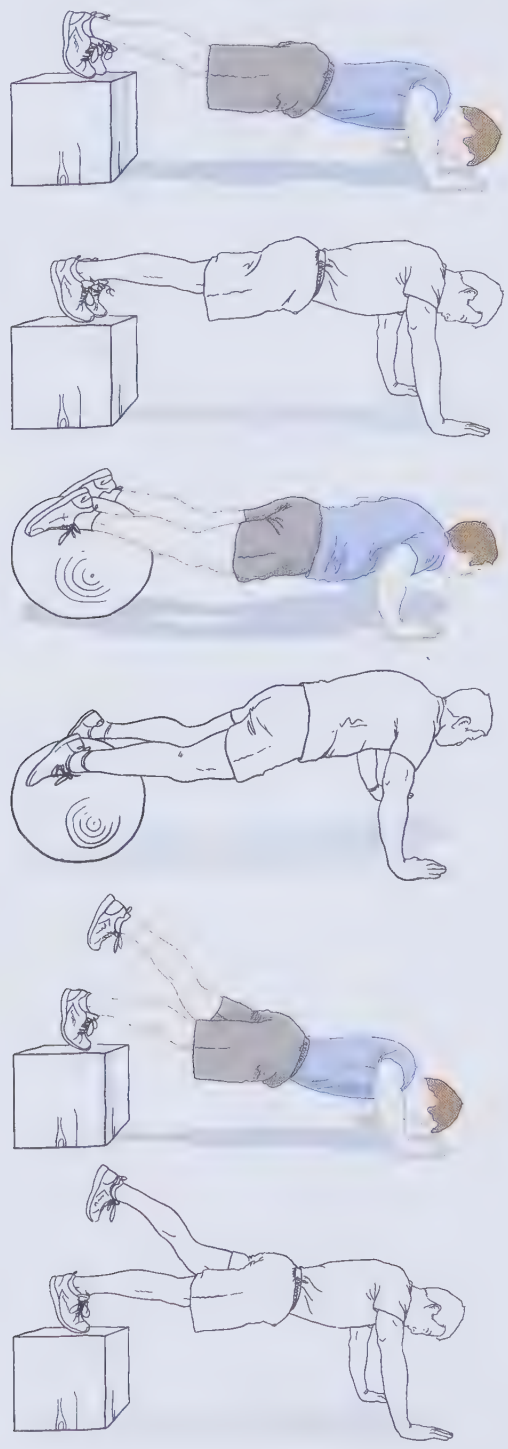


Fig. 7.14 Elevated Feet Push-Up

The above illustrations show variations on the standard push-up to make it more challenging. Once you can manage fifteen push-ups, consider trying these instead.

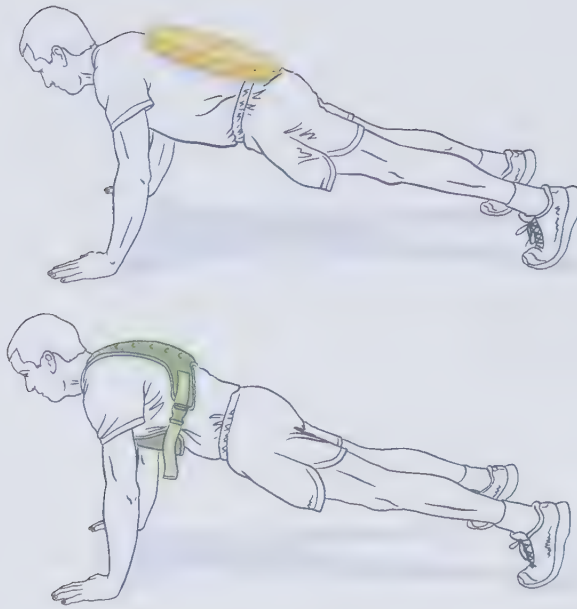


Fig. 7.15 Weighted Push-Up

A weight vest or weight plate are the best ways to increase push-up strength.

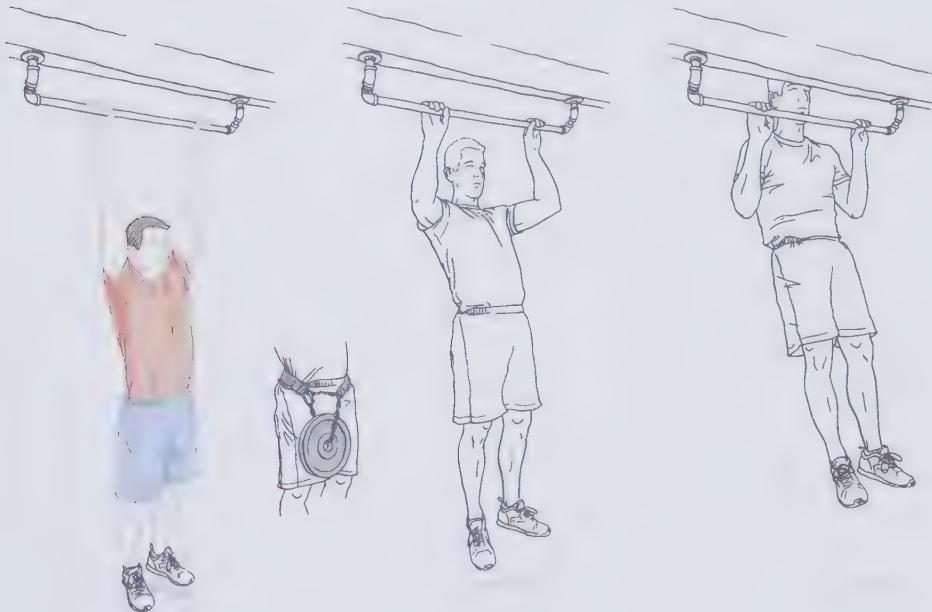


Fig. 7.16 Pull-Up/Chin-Up

The pull-up is the king of upper general body strength exercises and engages virtually every muscle from your waist up.



Mountain running demands a strong core. Here, Dan Patitucci grabs some meters after a summer snowstorm painted the meadows white. In the background is the Jungfrau, Switzerland. Photo: Janine Patitucci

Core Strength

Core strength is another aspect of general strength that plays a critically supportive role in locomotive sports like skimo and running. It serves this function by stabilizing the pelvis, spine, and shoulders so that the arms and legs can perform their propulsive functions most effectively. The stability of the pelvis is especially critical because it supports the bulk of the body weight, provides the anchoring for the legs, and provides the base of support for the spine. Many running injuries can be traced to weakness or imbalances of the core muscles, most of which attach to the pelvis. However, because of the general nature of core strength training, it is often overlooked by endurance athletes as unimportant. We caution against underestimating its usefulness. The core muscles are mainly made up of slow-twitch fibers. This imbues them naturally with a lot of endurance, but typically not so much strength. So, increasing their strength even a little can make a big difference for most athletes.

In both running and skiing, an effective arm swing causes a slight counterrotation of the shoulders relative to the hips, causing a little twist along the spine. If the core is not able to transfer this energy effectively from upper to lower body, you will be losing power.

Improving core strength is challenging because you first need to learn how to actively engage the deep muscle, transverse abdominis (TA), that is not used for locomotion but is the primary stabilizer of the pelvis. Because it doesn't move any major joints it is hard to isolate and engage, unlike the more obvious prime-mover muscles of the arms and legs. Learning to engage the transverse abdominis (see sidebar) is not like learning to do a bicep curl or a squat. So, it's hard to tell if it is activated.

Don't get discouraged by slow progress. It's normal. If you are having trouble holding good form and find your back swaying like an old horse or you can't hold the small of your back flat to the floor while doing leg lifts, you have poor pelvic stability caused by a weak core and can benefit from improving that strength.

How to Activate the Deep Core Muscles

This chapter's exercises require proper form to protect the spine and gain strength in the correct, deeper muscles. The TA is the deepest muscle in the core musculature and protects the spine and connects the upper and lower body. The TA wraps like a corset around your visceral organs. It ties together the pelvis and the lower six ribs in the front with the thoracolumbar fascia on your lower back to fully encircle the middle of your body. Unlike some of more easily activated core muscles like the rectus abdominis (seen in the six-pack) or the obliques, which flex the spine (like in a crunch) or twist the spine, the TA muscle is not used for movement. Its purpose is purely stabilization. When you engage the TA, it tightens against all its anchor points and provides stability to the lower spine and pelvis. But because this engagement does not result in noticeable movement, it can be hard to tell when you have activated the TA.

The easiest way to activate the transverse abdominis is by activating the pelvic floor muscle, which works simultaneously with the TA. To find the TA, lie supine on the floor with knees up and feet on the

floor and place your fingers on the ilia, the front hip bones. Move the fingers inward one inch (25 centimeters) and downward one inch. Pull up through your pelvic floor, as if stopping the flow of urine, and feel the muscle under each hand engage. Once familiar with this feeling, isolate the TA from the pelvic floor by drawing your navel to your spine, or imagine zipping up a tight jacket. Once you learn to activate the TA you can practice this while sitting or standing. If you feel your spine flex (like doing a crunch), then you are trying too hard and causing the activation of the rectus abdominis or oblique muscles.

When you have that feeling of tightening this girdle muscle, you can move to doing a plank exercise. In the plank position with forearms resting on the floor, there is no spine flexion (crunch) and the TA serves a key role in preventing the sagging of the hips. This is an isometric exercise, so you will just hold the position until you begin to shake or lose form (feel your hips sag or rise up as compensation). To maintain engagement throughout the workout, you may need to start your plank from the knees rather than from the toes.



Tamara Lunger, Philipp Reiter, and David Wallmann nearing the summit of Rheinwaldhorn in Switzerland. Photo: Mark Smiley

Core Strength Assessment

To test yourself to see if you have adequate core strength, do this simple test, which will allow you to most effectively target this critical strength.

PLANK

This exercise tests the strength of the transverse abdominis, the most important core stabilizer muscle. If you are deficient in the test, you will use the Stage 1 and 2 exercises explained below.

Get into the plank position as pictured with your weight on your toes and forearms, elbows directly under your shoulders and a neutral spine (no sag or arching). Time yourself while holding this position as long as you can before losing perfect form. Stop as soon as your back sags or your spine is no longer straight. Choose the stage of exercise that corresponds to your assessment results.

Stage 1 – Less than thirty seconds

Stage 2 – Thirty seconds to one minute

Stage 3 – Longer than one minute

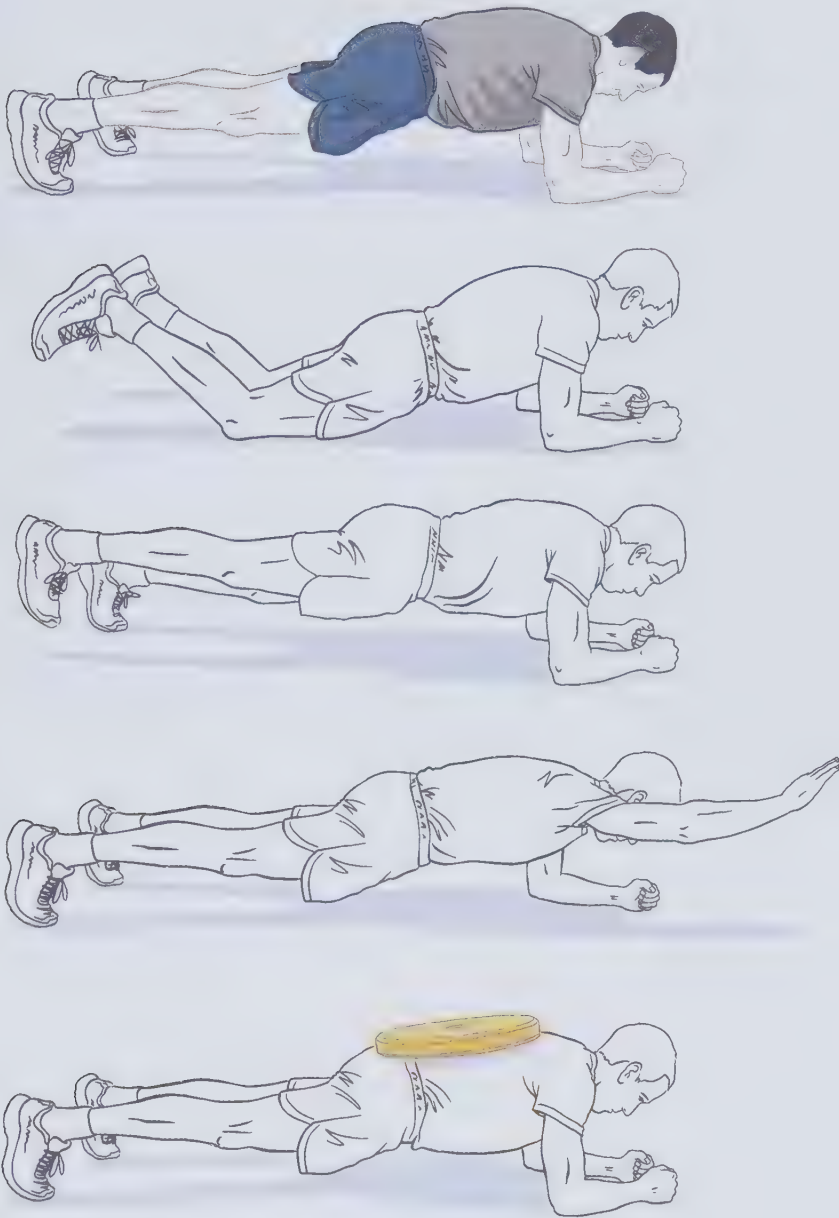


Fig. 7.17 Plank Variations

The top drawing shows good plank form. If you are unable to hold this position, then you can start from your knees. If you are stronger you can add challenge by lifting alternate legs and holding till you begin to lose form. Adding a weight or weight vest is a great way for the stronger athletes to improve transverse abdominal strength.





WINDSHIELD WIPER

This exercise tests the strength of the obliques, which are used for transferring power from the arms to the legs. The name for this one suggests the action of your legs as they wipe across an imaginary windshield.

Lie on your back with your arms outstretched from your shoulders with palms placed against the floor. Now flex your hips so that your feet are together with toes pointed at the ceiling. Slowly rotate your hips so that your feet lower to the floor to one side; keep your feet locked together with knees straight. You'll want to resist the rotation of your shoulders by pushing down hard with your hand on the side you are rotating toward. Just lightly touch the floor with the side of the lower foot before raising both feet back to the twelve o'clock position, and on to the other side where the other foot will touch down. Return your feet to the twelve o'clock position for one repetition.

Do this slowly and with control and perfect form. If you cannot manage to keep your knees straight, or legs together, then bend your knees and do the same rotation while keeping your knees together and knees pointed at the ceiling when you are in the twelve o'clock position. Choose the stage of exercise that corresponds to your assessment results.

Stage 1 – Zero reps with straight legs

Stage 2 – Fewer than five reps to each side with straight legs

Stage 3 – More than five reps to each side with straight legs

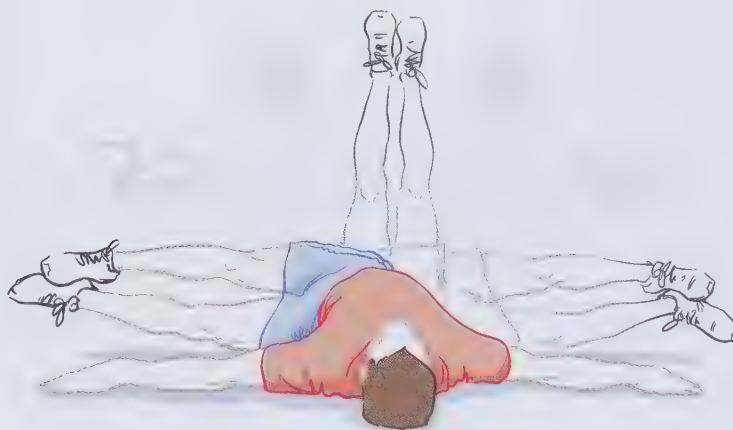


Fig. 7.18 Windshield Wiper

Lying with your hands outstretched to the sides at shoulder level and toes pointed to the ceiling swing your feet to almost touch the floor on each side, mimicking the action of a windshield wiper. Keep your knees straight, legs together, and toes pointed up. Move with control. Counteract your body's tendency to rotate with your arms. When you can do ten repetitions to each side, add challenge to this exercise by wearing boots or ankle weights.

Core Strength Workouts

Strict form is critical so that you target the weakest muscles in the chain rather than relying on stronger muscles, which can compensate for the weaker ones' inability to provide stability. If you feel your form break down, such as sagging hips, shaking, and wobbling, then you've overloaded that set of motor units and provided an adequate training stimulus.

We typically use the Core Strength routine as the finish to our warm-up before moving into the Leg Strength routine.

Stage 1

If you find yourself really struggling with either or both of these exercises, you need to develop more core strength. There is a good chance that you are simply unable to activate these core muscles effectively. To learn how to do this, you need practice. If you can't do these simple exercises, you will not be able to effectively use those muscles in your sport.

Exercises

Plank from the knees. 6x15 sec hold or until loss of form. Rest one minute between sets.

Windshield wiper with bent knees. Six sets of max reps. Rest one minute between sets.

Stage 2

If you can manage close to a minute of plank hold time without losing form or can do a few reps of windshield wiper with straight legs, then you'll benefit from performing longer sets of these as you prepare for Stage 3.

Exercises

Plank. Six sets of max holds until you lose form. Rest two minutes between sets.

Windshield wiper. Six sets of max reps. Rest two minutes between sets.

Stage 3

At this level it will be necessary to get creative and find ways of adding resistance to the exercises. Select three or four exercises from the list below. Feel free to find others that challenge the same muscles and movements. Use enough resistance so that you can manage no more than ten repetitions (total or per side) or thirty seconds of isometric hold before your form deteriorates from fatigue. Do no more than five sets of each exercise. Rest two minutes between sets.

Weighted Plank

Add a weight to rest on your butt. See above for information on how to do this exercise.

Weighted Windshield Wiper

Add weight to your feet by using ankle weights or heavier shoes or boots and perform straight-leg windshield wiper.

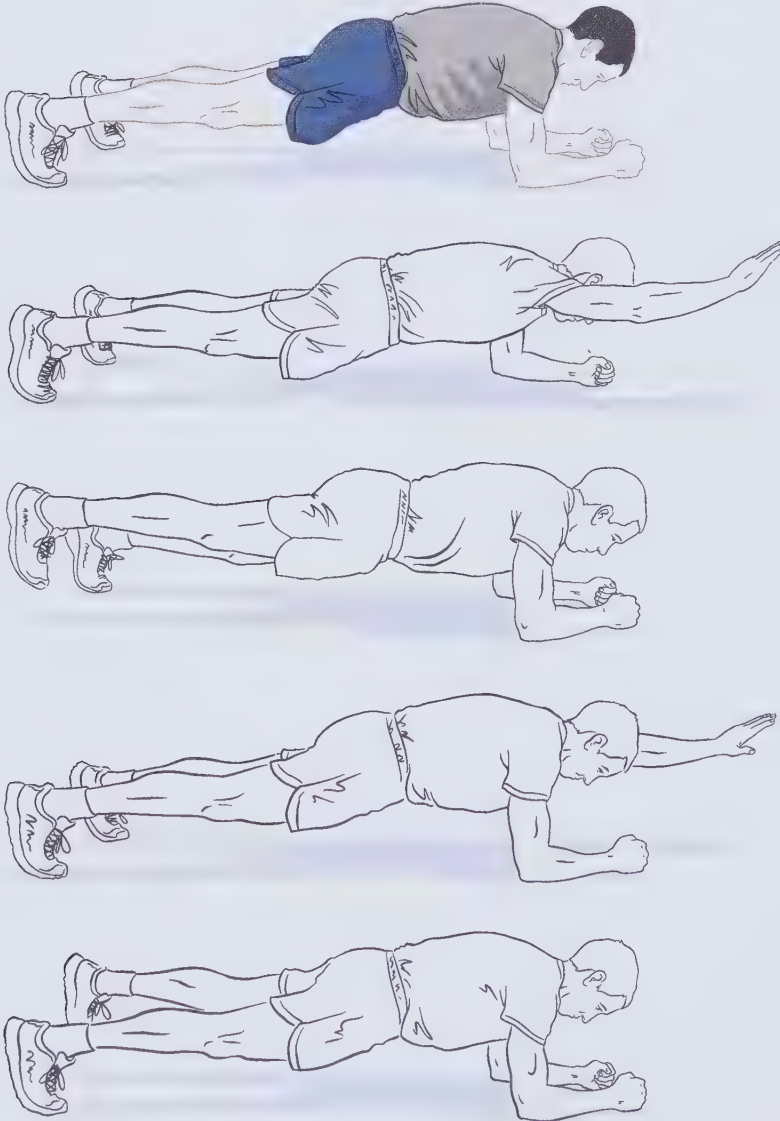


Fig. 7.19 Marching Plank

Lift each limb in succession. Hold until you start to lose form and then lift the next limb.

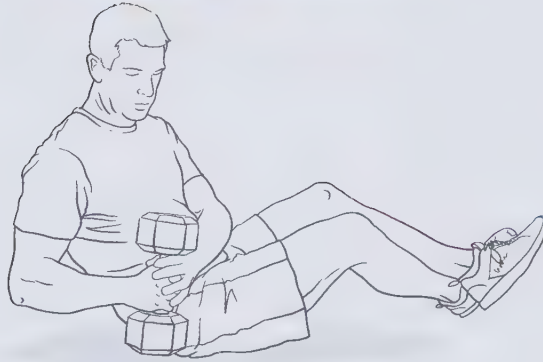
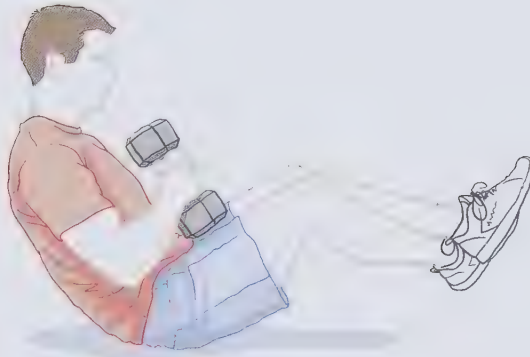


Fig. 7.20 Kayaker

Seated on the floor with feet off the ground and using a heavy dumbbell, rotate to touch the weight to the floor on each side.

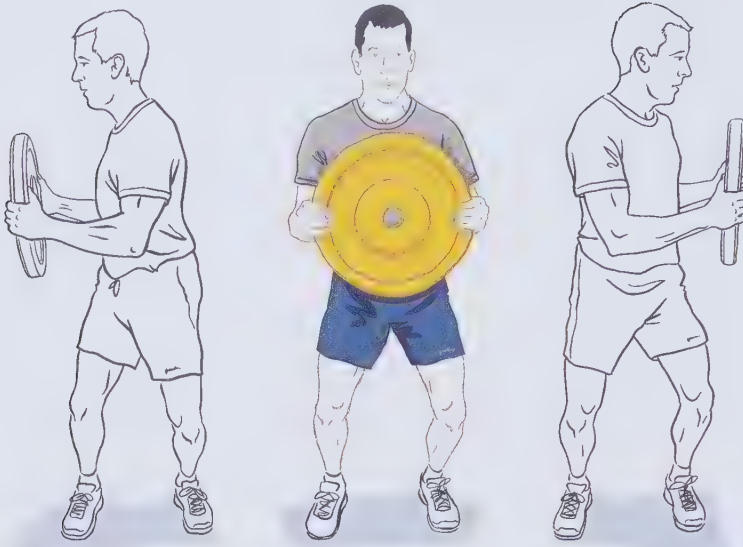


Fig. 7.21 Standing Trunk Rotation

With feet shoulder width and holding a weight in front of you, rotate aggressively in each direction. It is the stopping and starting of the rotation motion that provides the strength effect. Use a heavy weight and swing fast.

Planning Your General Strength Workouts Stages 1 Through 3

After the above assessments you should choose the appropriate stage you will want to start using for each of the leg, upper body, and core exercises.

1. Warm-up.
2. Core exercises. Complete the appropriate exercises for the stage you have determined.
3. Move to the main part of the workout (below) using the appropriate stage.

Stage 1 Workout

Four times through the following circuit with a two-minute rest between circuits:

- 1x20 reps: squats with knee band
- One set each of the appropriate stage push-up exercise (include appropriate stage pull-up exercise for skimo)
- 1x20 reps: single-leg lateral band

- One set of the appropriate stage push-up exercise (include appropriate stage pull-up exercise for skimo)

Stage 2 Workout

Three times through the following circuit:

- 1x15 reps per leg of box step-ups
- One set of appropriate stage push-up exercise (include appropriate stage pull-up exercise for skimo)
- 1x15 reps per leg of box step-downs
- One set of appropriate stage push-up exercise (include appropriate stage pull-up exercise for skimo)
- 1x15 reps of front lunge
- Rest three minutes and repeat

Stage 3 Workout

Four times through the following circuit:

- 1x6 reps of box step-ups with weight
- One set of appropriate stage push-up exercise (include appropriate stage pull-up exercise for skimo)
- 1x6 reps of box step-downs with weight
- One set of appropriate stage push-up exercise (include appropriate stage pull-up exercise for skimo)
- 1x6 reps of front lunge with weight
- Rest three minutes and repeat

ATHLETE STORIES

It Will Get Better

Anton Krupicka

One of the truisms of ultradistance events is that “it never always keeps getting worse.” The idea behind this is that no matter how physically and mentally depleted you may feel, as long as you persevere—put your head down, grit your teeth, and keep the faith—eventually your body will rebound from the trough that it’s in and you’ll experience a resurgence of energy.

For those who have never gone through this, it’s hard to believe and, indeed, illogical. It takes a degree of faith.

I had maybe my most profound experience with this quintessential low point followed by a high point in my very first mountain ultra: the 2006 Leadville 100. At mile 80, despite my pacer’s best efforts to goad me along, I was reduced to walking on flat ground, even downhill, and was pretty quickly doing mental calculations as to how long it would take me to walk the rest of the race. My legs had simply given up on the running motion.

However, only five miles later, I experienced an inexplicable surge in energy unlike anything I had felt since the first half of the race. In the matter of a few minutes, I went from despair to charging to the finish, running every step of the last fifteen miles, even occasionally putting my pacer into a spot of bother. At the time, I had no explanation, and I almost couldn’t believe that I was actually experiencing this illogical phenomenon that veteran runners had told me about before the race. Perhaps I just needed to smell the barn.

But there was one thing no one ever seemed to tell me: while an ebb in energy is almost guaranteed to happen at some point in the final quarter of these long efforts, the sneaky secret is the same thing can just as easily happen in the first quarter or third of a long race, too. And it doesn’t have to ruin your race if you don’t let it.

The very next year at Leadville, I was in great shape and leading the race 20 miles in when I gradually sunk into a perplexing funk. My energy and motivation

plummeted, and though I already held a commanding lead, I entertained a few insidious thoughts about dropping. How was I supposed to run another eighty miles at high altitude when I already felt this terrible? Fortunately, by backing off the intensity slightly, pushing the seemingly insurmountable remaining distance out of my mind, and eating and drinking a little extra, I gradually rebounded.

When I began to climb the race’s infamous Hope Pass at mile forty, I actually ran every step and went on to win the race in my personal best time, which remains the third-fastest time ever in the history of Leadville.

The point is, don’t get rattled if you feel off—or even downright awful—early in a long outing. The body’s peaks and valleys of energy are unpredictable and often hard to explain—especially when coming off of a long taper before a goal effort. But the thing to remember is that you are totally capable of rebounding and performing at your very best later on, no matter how improbable that may seem in the moment. Focus on pacing and staying present and positive. Don’t project too far into the day and despair at how much farther you have to go—and be diligent with nutrition and hydration—and things will eventually come around. ■

Anton Krupicka ran his first marathon when he was twelve years old. He is a two-time winner of the Leadville Trail 100 and a veteran of mountain ultraracing around the world. He used to spend all of his time going uphill on just his feet, but for much of the past decade he has put an equal amount of energy into going uphill on skis, a rope, and a bike, depending on the season. He lives in Boulder, Colorado.

Anton Krupicka training in Boulder, Colorado.
Photo: Fredrik Marmsater



